Robert C. Bridges

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PROFESSIONAL SUMMARY

Took an original spacecraft mission through SRR, SDR, and PDR, built a robotic bat from structure to circuitry to wing membrane, and simulated electron acceleration in the Van Allen Belt during a real solar storm. I am an aerospace and mechanical engineer (Cornell M.Eng. and B.S.) who carries a problem from the physics, through the model, to hardware on the bench. I want to solve the problem every flight program shares: ensuring the spacecraft flies and operates as designed, and proving it before launch.

EDUCATION

Cornell University, College of Engineering, Ithaca, NY

Master of Engineering, Aerospace Engineering — *May 2026*

Bachelor of Science, Mechanical Engineering — *Dec 2025*

Relevant Coursework: Spacecraft Technology and Systems; Introduction to Spaceflight Mechanics; Space Structures; Propulsion, Aircraft and Rockets; Mechanical Metallurgy; Combustion Processes; Plasmas for Propulsion

ENGINEERING PROJECTS

Original Spacecraft Mission Design, F.O.C.A. — *Jan - May 2026*

- Designed an individual spacecraft mission to intercept, capture, and analyze foreign objects in orbit, using prior flight missions as the baseline for the vehicle architecture so the design rested on flight heritage.
- Authored the SRR, SDR, and PDR documentation, defining requirements and validating the design against them at each review gate.
- Defined the full mission life timeline, working in Fusion 360, MATLAB, and Python as the sole engineer behind every system-level trade.

Robotic Bat Research, Master's Project — *Aug 2025 - May 2026*

- Designed and built a robotic bat, integrating its structure, circuitry, motors, and wing membrane into one working system and carrying each subsystem from concept through fabrication, test, and troubleshooting across mechanical, electrical, and materials domains.
- Fabricated and iterated structural parts in PLA, TPU, and resin, choosing rigid, flexible, or high-detail material and revising from test results.

Van Allen Belt Electron Simulation — *Jan - May 2024*

- Simulated electron acceleration in the outer Van Allen Belt during the May 2024 solar storm using leapfrog integration in a constant magnetic field, and benchmarked the results against observed energies up to 2.4 MeV.
- Simulated electron interaction with whistler-mode chorus waves and analyzed the results against the observed rise in radiation belt energy, physics that drives shielding, electronics, and lifetime margins for satellites in these orbits.
- Applied quantitative modeling to plasma physics and interpreted the gap to observed energies in light of effects the model left out.

Focus Glasses — *Aug - Dec 2025*

- Led market research and delivered a Product Design Specification with a 4-person team for glasses that keep users focused using invisible spectral flicker, translating user needs into engineering requirements.
- Completed finite element analysis in Ansys to ensure the safety of the glasses for the people who would wear them.
- Voted best new product in the class after explaining the technology to the public, professors, and students at an open exhibit.

Spin Wheel, Senior Design — *Jan - May 2025*

- Reinvented the computer scroll wheel with a touch screen and haptic feedback so people with disabilities could use it far more easily.
- Ran market research with the team to ground the design in real user needs, then built a prototype to test the concept in hand.
- Filed a patent application and presented at a public exhibition, taking one product from research through prototype to IP protection.

Chainsaw Dissection — *Aug - Dec 2024*

- Deconstructed a chainsaw to analyze its fluid dynamic properties, working backward from hardware to the engineering decisions behind it.

TECHNICAL SKILLS

- Spacecraft Systems: subsystems, SRR/SDR/PDR, requirements, mission timeline, spaceflight mechanics, propulsion, radiation belt physics
- Programming & Data Analysis: Python, MATLAB, Java (data analysis, modeling, numerical simulation, leapfrog integration, task automation)
- CAD & Design: Fusion 360, Ansys (FEA/CFD), GD&T
- Materials & Structures: composites (honeycomb sandwich panels, carbon fiber layups), structural analysis, mechanical metallurgy
- Fabrication & Tools: Red Apron Machine Shop, Micro Vu, 3D printing (PLA, TPU, resin), MS Office Suite, Adobe Creative Suite

PROFESSIONAL EXPERIENCE

Abbott Point of Care, Princeton, NJ

Project Engineering Intern — *May - Aug 2022 & 2023*

- Built and maintained a database of custom parts across Abbott's medical devices, streamlining search and documentation (2023).
- Analyzed bills of materials (BoM) and product structures for hundreds of components to support design and manufacturing decisions.
- Inspected circuit boards on Micro Vu precision measurement equipment; updated and calibrated TEAM, a company-wide error database (2022).
- Documented product issues and test results to support troubleshooting, issue investigation, and quality processes throughout the design cycle.

Princeton University, Facilities Department, Princeton, NJ

Engineering Intern — *May - Aug 2025 & 2026*

- Coordinated communication between project managers and on-site staff, producing daily progress reports to keep stakeholders aligned.
- Read and interpreted technical architectural plans and drafted a granite sign engraving template to support facility construction projects.

TEACHING, MENTORSHIP & CAMPUS INVOLVEMENT

- Teaching Assistant, Introduction to Spaceflight Mechanics & Introduction to Mechanical Design, Cornell MAE: graded assignments, led weekly labs, hosted office hours, and mentored students (2025 - 2026).
- Grader, Thermodynamics, Cornell MAE: created rubrics, graded homework, and delivered performance summaries to faculty (Aug - Dec 2024).

Involvement: Ultimate Frisbee, Social Media Chair & Captain (2022-2026) | New Student Orientation Leader (2022) | Mediocre Melodies, Member (2023-2024) | MEng Social Media Content Creator (2026)